

2G Enterprises

2G800 Automated Sample Handler System

User's Manual and Technical Reference

January, 1994

2G ENTERPRISES

2G800 Automated Sample Handler System

Table of Contents

1. Introduction	
1.1 Description of Equipment	1
1.2 Front and Rear Panel Descriptions of Control Console (2G810)	3
1.3 Front and Rear Panel Descriptions of Control Console (2G810)	5
1.3 Specifications	7
2. Operating Instructions	
2.1 Unpacking and Initial Setup	8
2.2 General Performance Check	8
2.3 Computer Protocol	10
2.4 Brief Descr. of Computer Commands and System Responses	11
3. Factory Switch Settings	14

Appendices

SMC25 User's Guide	Appendix A
List of Illustrations	2
Introduction	3
Pin Descriptions	4
Functional Description	8
Commands	10
Programming	16
Parallel Communications	18
Serial Communications	20
Motor Driven Connections	22
Inputs and Outputs	23
Jog Switches	24
Limit Switches	25
EEPROM Interface	26
Electrical Specifications	27
Physical Dimensions	28

Figures

1A Rear Panel of Control Console (2G810)	4
1B Front Panel of Control Console (2G810)	4
1B Front Panel of Control Console (2G811)	6
1A Rear Panel of Control Console (2G811)	6

Other figures	Appendix B
2 Block Diagram of Sample Handler System	1
3 Circuit Diagram of Half Step Driver Board	2
4 Circuit Layout of Half Step Driver Board	3
5 Circuit Diagram of Sample Handler Controller Board (2G810)	4
6 Circuit Layout of Sample Handler Controller Board (2G810)	5
7 Circuit Diagram of Sample Handler Controller Board (2G811)	6
8 Circuit Layout of Sample Handler Controller Board (2G811)	7

1. INTRODUCTION

1.1 Description of Equipment.

The 2G Automated Sample Handler System was designed to automate the sample handling of rock specimens in superconducting rock magnetometer systems. The unit can be mounted to all 2G rock magnetometers (e.g. models 2G760R, 2G750R and 2G740R). It can also be used on ScT rock magnetometers as an addition to the system or as a direct replacement of the manual sample handler.

The Automated Sample Handler System consists of a Sample Handler (2G801, 2G802, 2G803 or 2G804), a Sample Handler Controller (2G810 or 2G811) and a Sample Handler Drive (2G820). Several Sample Handlers are available; the selection of the appropriate system will depend upon whether an in-line degaussing system is used. The model 2G801 system is designed to be used without an in-line degaussing system. This system provides for rotation and approximately 1 meter of translation. Continuous rotation speeds of up to 5 Hz are possible to facilitate synchronous detection schemes designed to enable the measurement of extremely weak rock magnetism (total moments less than 10^{-8} emu).

The model 2G802 Sample Handler is designed to be used with an in-line 2 axis degaussing system, such as the model 2G600 Automatic Sample Degaussing System. The 2G802 is approximately 3 meters long and provides 2 meters of translation. A 2 axis (one axial and one transverse) degaussing system is mounted in-line with the sample handler so that rocks can be degaussed and moved back and forth between the degaussing and measurement regions automatically. The length of the degaussing system is about one meter (including mu metal shields). Three axis degaussing is accomplished sequentially by degaussing with the axial coil, translating to and degaussing with the transverse coil, then rotating 90° and again degaussing with the transverse coil.

The model 2G803 Sample Handler is designed to be used with an in-line single transverse degaussing coil. In order to achieve three axes of degaussing, a sample flip mechanism is included in the sample handler. This system has an overall length of 2.4 meters and a translation capability of 2.2 meters. The degaussing system including shields is about 0.4 meters long.

The model 2G804 Sample Handler is designed to enable measurements on long core rock specimens. This system consists of aluminum and fiberglass channels designed to support and guide the long core movement. The core itself is positioned in a nonmagnetic fiberglass carriage that is pulled in the channels by a pull rope attached to a geared high torque stepper motor. To enable precise measurement of the core position, two through beam sensors are used. These are interrupted by a flag attached to the sample handler carriage. When interrupted, the sample is defined as being in the "house" position. Two through beam sensors (typically located on either side of a rock magnetometer measurement system) are used to indicate a left moving and a right moving "home" position; this eliminates position errors stemming from stretch of the pull rope.

Movements of the Sample Handler are controlled by the 2G810 Sample Handler Controller for the 2G801, 2G802 and 2G803 Sample Handler. The 2G811 controller is designed to control movements of the 2G804 Long Core Sample Handler. Both of the 2G810 and 2G811 systems have an RS232 serial data link which enables communications with and control by an external computer. Sample movements are initiated by issuing high level commands over the serial data link. Sample movement parameters such as acceleration, velocity and movement interval are all user programmable.

The 2G810 Sample Handler Controller also provides two limit switches for sample movement (left and right) which stop the translation motor when activated and two home switches (translation and rotation). The 2G811 Controller also provides two limit switches for translation and two home switches (left moving and right moving). Two home switches are used to eliminate position errors due to pull cord stretch which occurs when long cores are moved. The 2G811 Controller does not provide for any type of sample rotation.

The 2G810 and 2G811 systems generate pulse strings to enable sample movements to take place. The pulse strings are fed to an appropriate 2G820 Stepper Motor Drive System. The 2G820 comes in two sizes, 1.2 Amp and 1.5 Amp. The rotation motors in the 2G801, 2G802 and 2G803 systems utilize a 1.2 Amp driver. The translation motors in these systems use a 1.5 Amp driver.

The stepper motor for long core systems utilizes a geared motor to achieve high torque. All other stepper motors are nong geared with 200 steps/revolution. For translation, one step produces a movement of about 0.65 mm. For rotation, one step provides a rotation of 3.6°.

1.2 Front and Rear Panel Descriptions of 2G810 Sample Handler Controller.

Front panel (see Fig. 1B)

Power switch: this switch turns on the main power (115V) to the system.

Power on LED: this indicator turns on when power is applied to the controller.

Jog Controls: these three push buttons control translation of the sample carriage. They are used to move the carriage in learning situations and when a computer controlled translation is not convenient.

Status: these LEDs indicate system status. When an external computer sends an attention command (@0,), the on line LED will turn on. When the translation and rotation driver consoles are properly connected to the controller console and powered up, the drive LEDs will turn on.

Translation Indicators: when the translation drive is selected (by sending 01,0, command) the translation selected LED will turn on. When the home, left or right limit switches are encountered, the appropriate LED will turn on.

Rotation Indicators: when the rotation drive is selected (by sending the 01,1, command), the rotation selected LED will turn on. When the rotation home switch is interrupted, the home LED will turn on.

Rear panel (see Fig. 1A)

RS232 Serial Output: a 25-pin D connector is provided to allow connection to a computer to control the sample handler system.

Home Switches Connectors: six-pin Bendix connectors are provided for the rotation and translation home switches. These are connected to the home switches on the sample carriage with cables supplied.

Limit Switches Connectors: three-pin Bendix connectors are provided for the left and right limit switches. These are connected to the limit switches on the sample handler with cables supplied.

Drives Connectors: six-pin Bendix connectors are provided for the rotation and translation drives. Supplied cables are used to connect to the driver consoles.

Fuse: wiring of the console is protected by an easily replaceable 1 Ampere fuse.

Power connection: 115V power is connected to the console using the supplied power cord.

Fig. 1A Rear Panel of Sample Handler Controller

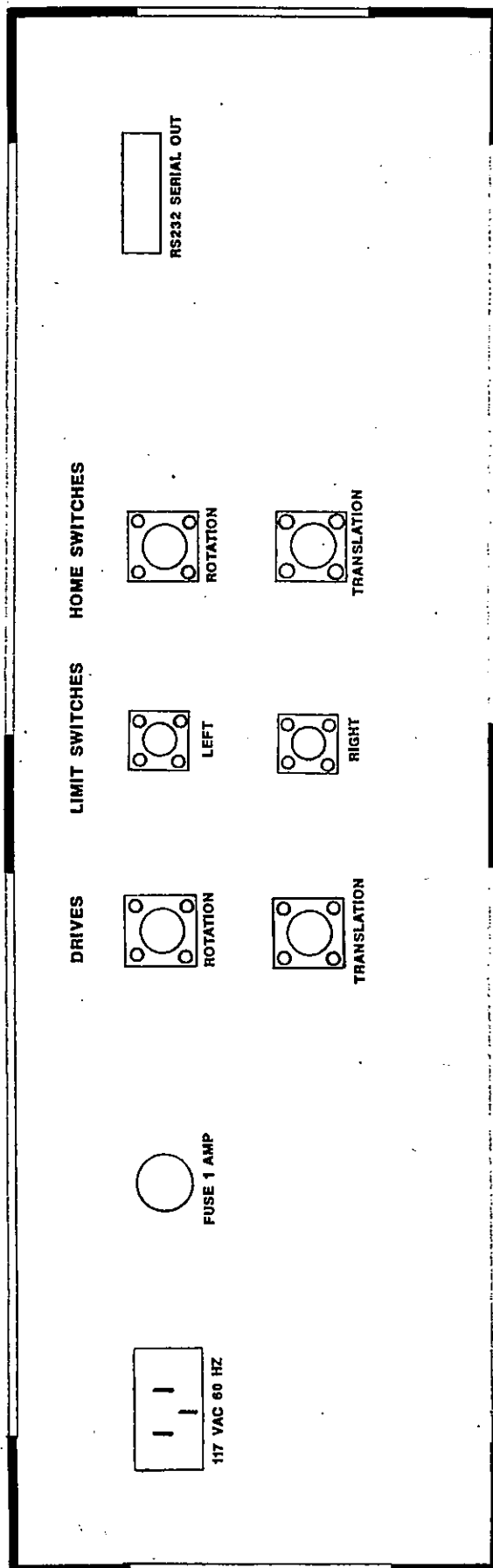
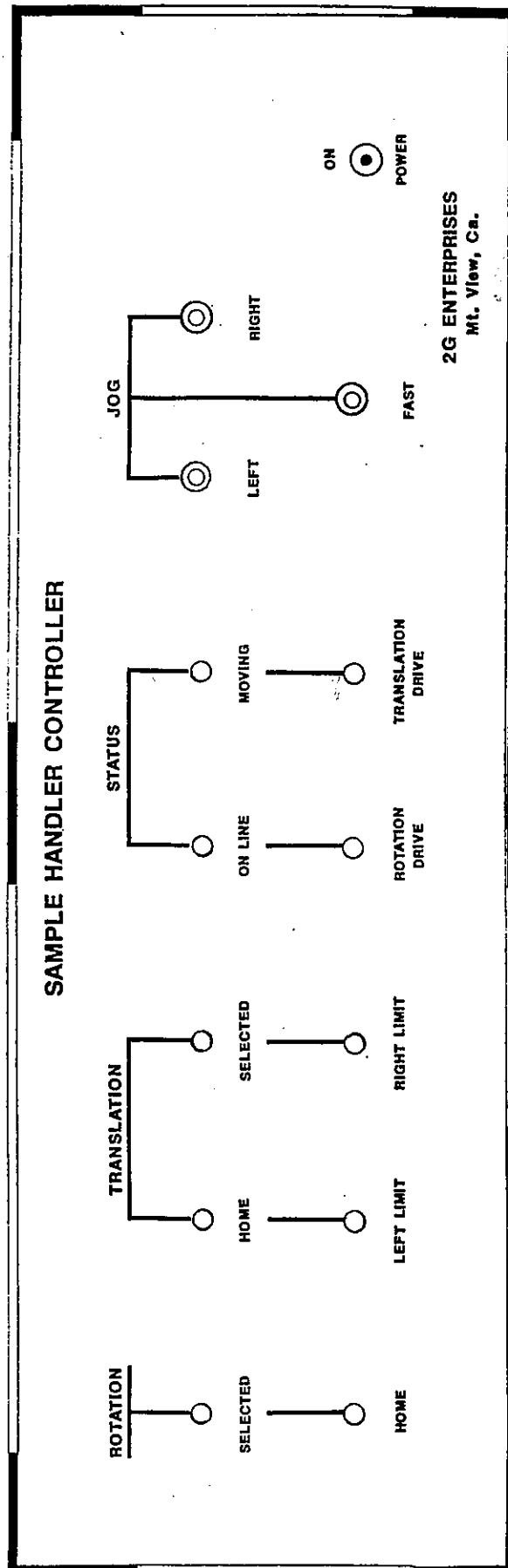


Fig. 1B Front Panel of Sample Handler Controller



1.3 Front and Rear Panel Descriptions of 2G811 Sample Handler Controller.

Front panel (see Fig. 1C)

Power switch: this switch turns on the main power (115V) to the system.

Power on LED: this indicator turns on when power is applied to the controller.

Jog Controls: these three push buttons control translation of the long core carriage. They are used to move the carriage in learning situations and when a computer controlled translation is not convenient.

Status: these LEDs indicate system status. When an external computer sends an attention command (@0,), the on line LED will turn on. When the translation driver console is properly connected to the controller console and powered up, the drive ready LEDs will turn on.

Left Limit and Right Limit: These LEDs turn on when the left and right limit switches are activated. This occurs when the sample carriage moves to the left or right end of the sample handler. Power is interrupted to the stepper motor.

Left and Right Home: these LEDs are normally green when the position sensing beam is uninterrupted. When interrupted, they turn red, indicating that the long core carriage is in the home position.

Moving: this LED is illuminated whenever the sample is translating.

Rear panel (see Fig. 1D)

RS232 Serial Output: a 25-pin D connector is provided to allow connection to a computer to control the sample handler system.

Home Switches Connectors: six-pin Bendix connectors are provided for the left and right home switches. These are connected to the home switches on the sample handler with cables supplied.

Limit Switches Connectors: three-pin Bendix connectors are provided for the left and right limit switches. These are connected to the limit switches on the sample handler with cables supplied.

Drives Connectors: a six-pin Bendix connector is provided to enable connection to the translation drive. Supplied cable is used to connect to the driver console.

Fuse: wiring of the console is protected by an easily replaceable 1 Ampere fuse.

Power connection: 115V power is connected to the console using the supplied power cord.

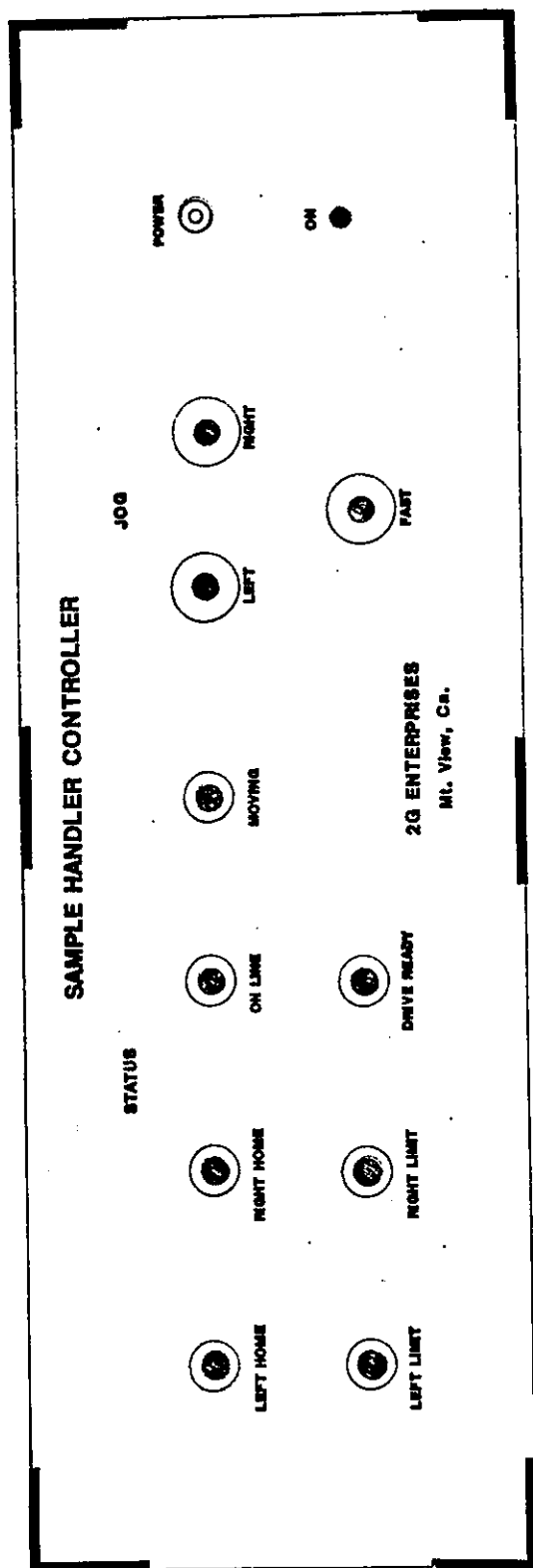


Fig. 1C Front Panel of Sample Handler Controller (2G811)

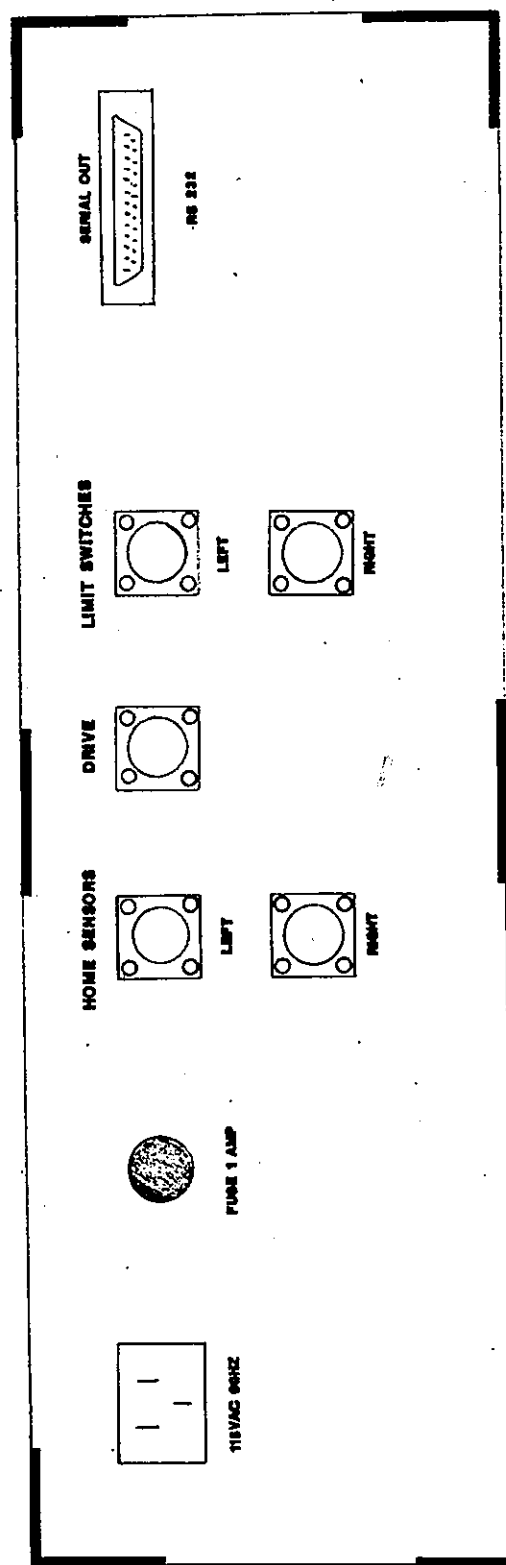


Fig. 1D Rear Panel of Sample Handler Controller (2G811)

2. OPERATING INSTRUCTIONS

2.1 Unpacking and Initial Setup.

Unpack the instrument; if you find any damage, report this to 2G Enterprises and the carrier responsible for delivery.

The sample handler system consists of the following components:

1. Controller console
2. Translation driver console
3. Rotation driver console
4. Sample carriage
5. Interconnect cables

To set up the instrument, place the various parts of the system in a convenient location and connect all of the interconnection cables.

Connect the controller console to the two driver consoles using the cables with 6-pin Bendix connectors on each end. Connect the translation drive cable to the large translation driver console and the rotation drive cable to the small rotation driver console. Do not interchange these cables; driving the rotation stepper motor with the large translation driver console may damage the motor.

Connect the left and right limit switches to the control console. The left limit switch is the one nearest to the translation stepper motor. The right limit switch is nearest to the magnetometer system.

Connect the rotation and translation home switches to the control console. Distinguish these connectors by following the cable back to the home switch locations on the sample carriage.

Connect an external computer to the 25-pin D connector on the controller console.

2.2 General Performance Check.

After making all interconnects, turn on the power switches on the control console and translation and rotation drivers. The power indicator lights and the translation selected LED should come on. The rotation and translation drive LEDs should also be on indicating proper cabling and powering of the driver consoles. Simultaneously push the right and fast jog push buttons. The translation stepper motor will move the sample carriage to the right. Translate the carriage until the right limit switch is encountered. Jogging will stop at this point and the right limit LED will turn on. Press The left and fast jog push button and jog left until the translation home sensor is encountered; the translator home LED will turn on. Continue jogging to the left until the left limit switch is encountered, turning on the left limit switch LED.

To continue with system checkout it will be necessary to send computer commands. Connect an external computer to the 25-pin D connector on the control console rear panel.

Send the following character string to the sample handler system:

@0, [0=zero]

When this is received by the system the on-line LED will turn on. Next send the command:

+,A20,D20,M12000,N20000,G,

This command sets the acceleration and deceleration at 20, the velocity at 1200 steps per second, the distance at 2000 pulses and instructs the system to execute a programmed move. The system should translate away from the translation motor a distance corresponding to 2000 pulses. If a limit switch is encountered the system will stop. When a limit switch is encountered it is necessary to issue a reverse direction command (+ or -) before any other move commands are issued. Immediate translation can be initiated by issuing an S, to start translation and a Q, to quit translation.

Further Commands for the 2G810 System:

To check the seek home command send H1. The motor will translate the system to the home sensor and decelerate to a stop. The motor will then slowly jog back until the home sensor is reencountered.

To check rotation, send the command:

01,1 [0=capital letter O]

This redirects output to the rotation system. Rotate continuously by sending S, and quit by sending Q,. Programmed rotations can be achieved by issuing the G command. Previously set up parameters for acceleration, velocity, etc. are not lost when changing between rotation and translation. Check the rotation home sensor by issuing the H1 command.

Because stepper motor driven systems move in discrete steps, certain velocities and accelerations are smoother than others. In general, these parameters should be selected to avoid exciting the natural resonances of the system. We find that velocities near the top end (M12000) and accelerations in the 15(high) to 30(lower) area are most suitable. If accelerations are set too high the system may lose steps and accuracy of movement will be impaired.

Further Commands for the 2G811 System:

The two home switches used in long core (2G811) systems must be individually activated, i.e. they cannot both be active. To activate the home switch 1 issue the command:

01,0 [0=capital letter O, 0=zero]
(output 1 to the 0 state)

To activate home switch 2, issue the command:

01,1 [O=capital letter O, 0=zero]
(output 1 to the 1 state)

Once the proper home switch has been activate, an H1 command can be issued to enable the system to seek home.

Sensitivity Adjustment for Through Beam Sensors (2G811 System only)

The home sensor may be adjusted for sensitivity. This may be needed due to the amount of ambient light present. After mechanical alignment of the photo sensors is done, remove the top cover from the Sample Handler Controller console. While watching the front panel LED for the right home switch, adjust R10 on the main control board to get desired sensitivity. Adjust R11 to get the desired sensitivity on the left home switch.

Step Rate Select Switch (SW4) (2G811 System only)

The Sample Handler Controller has an adjustable step rate switch. This switch engages a divide-by-10 counter. For normal operation, this switch should be in the position towards the front panel, which enables the counter. If the Controller is used with a micro step drive, the switch should be moved to the position towards the rear panel. This deactivates the divide-by-10 counter.

2.3 Computer Protocol.

All communications between the 2G800 Sample Handler Systems and external computers are achieved using the RS232 serial data link. The data link connector on the system is a 25-pin female "D" connector. External computer commands are transmitted on pin 2 of this connector. The Sample Handler System transmits replies to the external computer on pin 3. Pin 7 is the ground. Pin 5 of the connector is a busy signal which, when high, indicates that the system is busy processing commands or sending data. For most applications, only pins 2, 3 and 7 need to be connected to the external computer.

The main signal processing circuit used in the 2G810 and 2G811 System is a single chip processor designated SMC25 and manufactured by Anaheim Automation, Inc. A detailed manual describing the operating characteristics of this processor is included in Appendix A of this manual. In section 2.4 of this manual, we also provide a quick synopsis of commands that are often used.

At the factory, the system serial data link protocol is set up for 1200 baud, 8 data bits, no parity and one stop bit. Switches on the printed circuit board of the control chassis unit enable 300, 9600 and 76,800 baud communication rates. The address of the system is factory selected to be 0. This can be changed to any value between 0 and 31. Refer to section 3, Factory Switch Settings, for instructions on changing baud rate or system address.

2.4 Brief Description of Computer Commands and System Responses.

1. Establishing Communications. Before any commands can be sent to the 2G800 System, the system must be put on line. This is accomplished by sending the character string

@0, [0=zero]

When the system is on line, the "on line" status LED located on the front panel will be on. This command will work if the system address is the factory setting of 0 (zero). Substitute the system address for the 0 if this is changed. Note that all commands are terminated with a comma (,) which is also ASCII character 44.

2. Setting Up Acceleration and Deceleration. These commands consist of either an A (Acceleration) or D (Deceleration) followed by a decimal number between 0 and 127. Highest acceleration and deceleration is 0 and lowest is 127. If acceleration or deceleration is set too high, the system may miss steps and accuracy will be degraded. Settings in the 20-50 range are usually employed. Power on default is 5 for acceleration and 10 for deceleration. Examples of acceleration and deceleration commands are:

A20,
D15,
A50,

3. Setting Up Velocity. This command consists of an M followed by a decimal number between 50 and 12,000, for example

M1200,

Do not exceed 12,000 for this parameter or erratic behavior will result. The decimal number issued is 10 times the actual pulse rate to the motor. Since the motor requires 200 pulses (full step) or 400 pulses (half step) per revolution, a speed setting of M10000 sets the motor to revolve at 5 revolutions per second in full step or 2.5 revolutions per second in half step. This rate is one-half the sample rate rotation due to the pulley ratios. The sample handler is set up at the factory for half stepping.

4. Setting Up the Distance to be Traveled. The translation or rotation distance is set up by issuing an N command followed by a decimal number between 1 and 16,777,215. The actual number of pulses supplied to the motor is the decimal number divided by 10. For this reason, only numbers evenly divisible by 10 should be used after the N command. Use of numbers not divisible by 10 will result in sample position creep. Full scale travel on a 53 inch long (travel) system is about 42,000 counts (half stepping). One sample cycle (360°) or rotation is 2000 counts (half stepping).

5. Motor Direction. This is set by issuing either of the following commands:

- +, translation into rock magnetometer or clockwise sample rotation
- , Translation out of rock magnetometer or counterclockwise sample rotation

6. Axis Selected.(2G810 System only) In the two axis system (translation and rotation), translation mode is selected by issuing the command:

01,0 [O=capital letter O, 0=zero]

To select rotation, issue the command:

01,1 [O=capital letter O]

The translation or rotation selected LED on the front panel will turn on in accordance with which mode is selected. When the unit is first turned on, the default mode is translation.

7. Home Switch Selected.(2G811 System only) To activate home switch 1. send the command:

01,0 [O=capital letter O, 0=zero]

To activate home switch 2, send the command:

01,1 [O=capital letter O]

8. Movement Commands. Two immediate commands are available:

- S, start continuous motion using acceleration and velocity parameter
- Q, Stop motion using deceleration parameter

Programmed movements (for a given number of counts specified by the N parameter) are initiated by the G, command. Other examples of movement commands are:

+G,
+S,
-G,

8. Verification of Parameters. Verification of stored motion parameters can be accomplished by issuing a V command followed by a letter to indicate what variable is to be verified. For example:

VP, verify position (internal counter)
VN, verify programmed move distance
VA, verify acceleration parameter
VM, verify velocity parameter

When interrogated in this way, the system will respond by sending back ASCII characters for the values requested, followed by a carriage return to terminate the reply.

9. Verification of System Status. To determine the system status, issue the following command:

%,

Upon receiving this command, the system will answer back with a single character (with no line feed or carriage return character following). The returned character denotes the system status as follows:

5 end of move, previous G command complete
7 hard limit stop
G motor is currently indexing

Other responses are described in the Anaheim Automation manual included in Appendix A of this manual.

If a command

GF%,

is issued, the imbedded F has the effect of inhibiting the return status character until the motor stops indexing. This is a useful option to use when an interrupt is desirable at the end of a programmed move.

10. Sample Flip. If the system has sample flip capability, this is initiated by issuing the command

02,2, [O=capital letter O]

Return the sample to the unflipped position with the command

02,0, [O=capital letter O, 0=zero]

11. Seek Home Command. The system is equipped with both rotation and translation home sensors. These sensors allow the system to start from a known position at initial power on. The command format is:

H1,

Note that if the direction is set at + when the carriage is on the rock magnetometer side of the translation home sensor, the carriage will never encounter the sensor; interruption of the right limit switch will stop the motion in this case.

3. FACTORY SWITCH SETTINGS

The sample handler is set up at the factory as follows:

1. 1200 Baud data rate
2. Half stepping mode
3. Address 0

These can be reconfigured if desired. To change the mode between 1/2 step (400 steps/rev) and full step (200 steps/rev), remove the top cover on the translation and rotation driver consoles and change the position of the slide switch mounted on the printed circuit board. Select 1/2 step mode by moving slide switch to position nearest the edge of the printed circuit board.

To change the system address, remove the controller console top cover. The system baud rate is determined by the position of the switches on dip switch SP1 near the edge of the printed circuit board. Switch positions 6, 7 and 8 select baud rate as follows

<u>Baud</u>	<u>6</u>	<u>7</u>	<u>8</u>
300	off	off	on
1200	on	off	on
9600	off	on	on

The system address is determined by the setting of switch positions 1 through 5. Switch position 1 is the least significant bit and position 5 is the most significant bit. The system address is binary coded.

APPENDIX A

SMC25

USER'S GUIDE

ANAHEIM AUTOMATION

910 E. Orangefair Lane

Anaheim, CA 92801

(714) 992-6990

Copyright 1987 by Anaheim Automation

April 1987

Telex: 2978217 MCI

Fax: (714) 992-0471

Table of Contents

	Page #
List of Illustrations	2
Introduction	3
Pin Descriptions	4
Functional Description	8
Commands	10
Programming	16
Parallel Communications	18
Serial Communications	20
Motor Driver Connections	22
Inputs and Outputs	23
Jog Switches	24
Limit Switches	25
EEPROM Interface	26
Electrical Specifications	27
Physical Dimensions	28

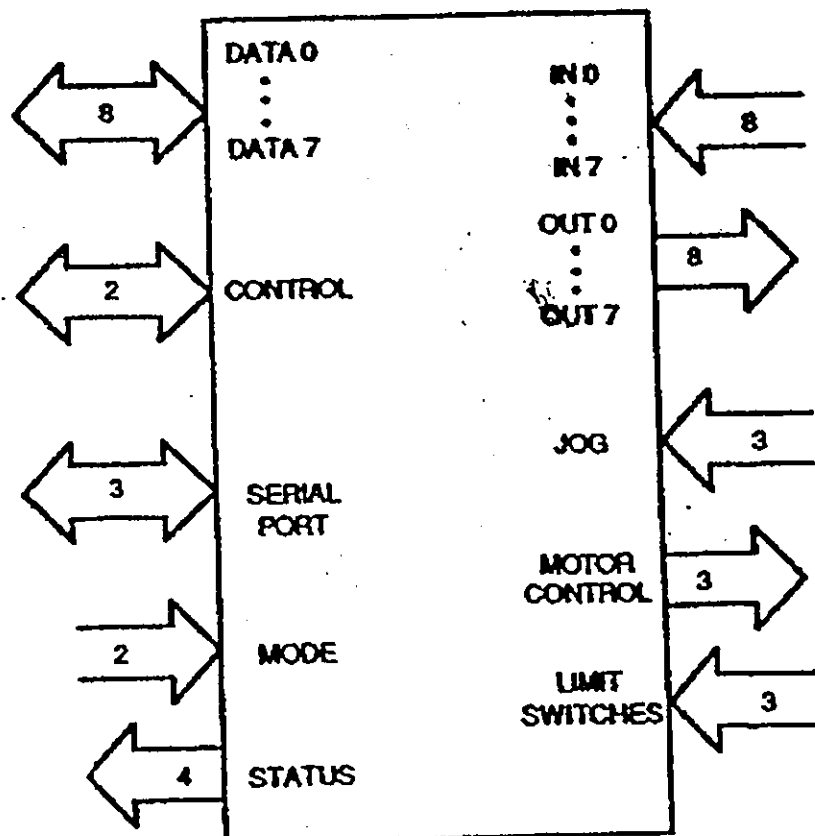
List of Illustrations

	Page #
Chip Diagram	3
Pin Assignment	4
Memory Usage Table	17
Parallel interface example	18
Parallel transfer timing diagrams	19
Serial interface example	20
Baud rate table	21
Motor Driver Example	22
Parallel address input	23
Jog switch example	24
Limit Switch Example	25
EEPROM Connection Diagram	26
DC Characteristics	27
AC Characteristics	27
Absolute Maximum Ratings	27
Physical Dimensions	28

Introduction

The SMC25 is a high performance single chip solution to step motor control allowing easy control of speed, acceleration, distance, and general purpose I/O pins. Provisions have been made for control through either an asynchronous serial port or a parallel data port with handshake controls and status lines. The user may write programs to store and execute within the SMC25 and these may be saved in an external serial EEPROM to be loaded and executed automatically on power up.

Some of the significant improvements that the SMC25 incorporates over previous step motor control chips are 24 bit position counter, communications while the motor is running, and speed change while the motor is moving. A large program area, many I/O lines, and a very rich instruction set combine to allow the user to write programs which in many cases may not even require a host controller in the system after initial program development.



Block Diagram

Pin Descriptions

The SMC25 is packaged in a 64 pin shrink dip. This package provides the advantage of a large number of I/O pins without the expense of a pin grid array or the large size of the normal 64 pin dip. The spacing between pins is set at .070 inch as opposed to the normal .100 inch and the row spacing is .750 inch versus .900 inch to provide an overall size only a little larger than a normal 40 pin dip.

Vcc All pins with this designation should be connected to the positive side of the power supply.

Vss All pins with this designation should be connected to the negative or ground side of the power supply.

DATA 0-7 In the parallel mode of operation these pins are an eight bit bidirectional data bus over which all commands and data are transferred. In the serial mode these pins select the baud rate and the address of the chip. DATA 0-4 provide for one of thirty-two addresses and DATA 5-7 select the source and rate of the serial clock.

SMC25

1	Vss	14 XTAL	64
2	XTAL	EEPROM DI	63
3	EXTAL	EEPROM SK	62
4	Vcc	EEPROM CS	61
5	Vcc	INST/DATA	60
6	RESET/	INPROG	59
7	Vcc	DATA 0	58
8	Vcc	DATA 1	57
9	Vcc	DATA 2	56
10	STEP	DATA 3	55
11	SERIAL CLK	DATA 4	54
12	RX DATA	DATA 5	53
13	TX DATA	DATA 6	52
14	EEPROM DO	DATA 7	51
15	REQ	BUSY	50
16	RAMP CLK	DIR	49
17	IN 0	SRO	48
18	IN 1	SELECTED	47
19	IN 2	STEP DIR	46
20	IN 3	POWER	45
21	IN 4	RUNNING	44
22	IN 5	MPX	43
23	IN 6	Vss	42
24	IN 7	OUT 0	41
25	PARSER	OUT 1	40
26	8/4.9 MHz	OUT 2	39
27	FAST	OUT 3	38
28	JOG +	OUT 4	37
29	JOG -	OUT 5	36
30	HOMELS	OUT 6	35
31	SOFTLS	OUT 7	34
32	HARDLS	Vcc	33

Pin Assignment

REQ	This signal is driven by the host controller. In the parallel mode if DIR is low then a low on this pin indicates that the host has placed data on the data bus. If DIR is high then a low on this pin indicates that the host is ready for the SMC25 to place data on the data bus. In the serial mode a low on this signal indicates that the host is able to accept data from the SMC25 and a high level will prevent the SMC25 from sending any data. If the SMC25 has data to send it will wait for this signal to be low before send or accepting any data.
BUSY	Handshake signal indicating the SMC25 is busy processing or sending data. For a more detailed description see the sections on communications.
DIR	Output showing the direction the SMC25 expects data to travel on the data bus. A <u>high signal</u> indicates the SMC25 is sending data to the host while a <u>low signal</u> indicates the host should be sending data to the SMC25.
SRQ	Service request output. A <u>high level</u> on this signal indicates that the SMC25 has a message to send to the host. This is not used for the verify commands but for unexpected messages such as errors or end of index.
PAR/SER	A mode select input to determine whether the SMC25 communicates over the parallel port (a high level) or the the serial port (a low level). The effects of this pin will be discussed in further detail in the communications section.
8/4.9 Mhz	A mode select input to tell the SMC25 the value of the crystal being used. This value may be subsequently over written by a load from the EEPROM or a command from the host. A high level selects 8 megahertz and a low level selects 4.9152 megahertz.
HARD LS	When the hard limit switch input is low the motor will not step. If it was moving at the time the line changed to low then it will stop without a ramp.
SOFT LS	When the soft limit switch input goes low the motor will ramp down and complete the move at base speed. If it is low when a move command is received the motor will move but will not ramp up.

HOME LS	The home limit switch input is used by the home command to stop the motor. It has no effect at any other time.
JOG +	When this line is low and the motor is not executing a programmed move the motor will run in the positive direction at either the base speed or the jog rate depending on the state of the FAST signal.
JOG -	The same as JOG + except that the direction is negative.
FAST	A high level on this input selects jogs to be done at the slow rate while a low selects the base speed for jogs.
SELECTED	This output is low when this is the most recently selected device. It will go high if another device is selected or all devices are deselected.
IN 0-7	A general purpose input port.
OUT 0-7	A general purpose output port.
TX DATA	Asynchronous serial port output. Used in the serial communications mode.
RX DATA	Asynchronous serial port input. Used in the serial communications mode.
SERIAL CLK	Bidirectional line for the BAUD rate of the serial port. If the chip is programmed for an external BAUD rate then this line is an input, otherwise it is an output of the selected rate.
RESET/	Master reset signal for the SMC25. This signal is active low.
XTAL	Crystal control output. If an external oscillator is used this pin should be left open.
EXTAL	Crystal control or external oscillator input.
POWER	This output is intended to control the power to the motor. It will be low, power on, whenever the motor is moving. It will be set high, power off, at some time after the motor stops depending on the programmed hold time.
STEP DIR	The direction of the motor is controlled by this output. A high level is used for the positive direction.

| = +

STEP OUT	This output is the step clock for the motor. For normal operation it should be connected to the RAMP CLK input directly and to the motor driver through an inverter.
RAMP CLK	The ramp clock input controls the ramping of the motor speed. This will normally be connected to the STEP OUT signal directly, but the user may use external circuitry to develop custom ramps.
RUNNING	A status output that is high whenever the motor is indexing. It is low at all other times.
INPROG	This output is high when the SMC25 is executing an internal user program.
INST/DATA	This status output is low when the SMC25 expects the next communication to be an instruction character and high when it expects a data character.
MPX	In parallel mode this output will multiplex the input pins IN 0-3 with the address of this device. A high level selects the address and a low level selects the inputs. If the inputs are not being used then the address may be connected directly without the multiplexer. This output has no use in the serial mode of operation.
EEPROM DI	This pin connects to the DI pin of the EEPROM.
EEPROM SK	This pin connects to the SK pin of the EEPROM.
EEPROM CS	This pin connects to the CS pin of the EEPROM.
EEPROM DO	This pin connects to the DO pin of the EEPROM.
1/4 XTAL	System clock output equal to one quarter of the crystal frequency.

Functional Description

The SMC25 accepts high level commands from the user and converts them into clock pulses and other control signals needed to operate a step motor. The command structure is designed to be easy for both a user at a dumb terminal and for control from a computer program.

Host Interface

The host interface can be either serial or parallel. The serial mode of operation is intended primarily for situations where the SMC25 is external to the host system. The parallel mode is intended for use on boards within a bus oriented computer system.

The serial interface is a standard asynchronous communications port, compatible with most terminals or computer systems. It need only be connected to the proper buffers for RS232 or RS422 to connection to the user's system. The BAUD rate can be selected as internal or external. If internal is selected then the serial clock pin becomes an input. If the internal generator is selected then there are four divisors available of the system clock. If the crystal is 4.9152 Mhz then these will give normal BAUD rates of 300,1200,9600, and 76800. The busy and request lines are used as handshake controls, busy for the SMC25 to hold off the host from sending, and request for the host to hold off the SMC25 from sending.

The parallel mode provides the most robust interface. There are numerous status lines brought out of the chip to aid the user in knowing what the SMC25 is doing. Because of this and the speed advantage of parallel operation it is the preferred mode. A minimum interface would only need to use REQ, BUSY, and the DATA lines. Additional status lines to be monitored are, in order of preference, DIR, SRQ, Inst/Data, Running, Inprog, and Selected.

Motor Interface

The motor interface is four lines, STEP DIR, STEP, RAMP CLK, and POWER. Step should always be connected to the ramp clock pin. The power pin controls the all phases off input offered on may drivers. STEP and STEP DIR control a driver that accepts clock and direction inputs.

Inputs and Outputs

The general purpose inputs and outputs, of which there are eight of each, are provided for the user for application specific switches and solenoids, or whatever else can be controlled from the parallel port of a processor. The user may access these pins anytime the SMC25 is not executing an internal program.

Limit Switches

There are three kinds of limit switches available. The hard limit switch stop all pulses to the motor immediately. The soft limit switch will cause the motor to ramp down and complete its move at the base speed. The home limit switch is used only by the home command to locate the home position. These inputs should be externally multiplexed by the step direction line to separate the positive limit from the negative limit, otherwise, the motor will not be able to move away from a hard limit.

EEPROM Interface

The SMC25 provides an interface to a serial EEPROM manufactured by National Semiconductor. This external memory is used to store the internal program and a few parameters for recovery after a power failure or the machine being turned off. The user may also set the internal program to automatically start executing on power up.

Jog Switches

The jog inputs allow manual control of the motor while maintaining the internal absolute position counter. This is important for implementing a teach mode of operation at the system level. The user can manually select between two speeds: the slow jog rate or the base speed, by use of the FAST input.

Commands

Command Format

Commands are made up of one or two letters or a letter and one, two, or three numbers. All letters are upper case 8-bit ASCII characters with the most significant bit set to zero. All numbers are decimal ASCII sent most significant digit first followed by a nondecimal character. Spaces have no effect on the command format. The character that terminates a number will be used as the next command. For this reason a comma, carriage return, or line feed are accepted as null commands, not changing anything and not giving a command error.

Command Summary

A zz

Acceleration is a number from 0 to 127. This controls the rate at which the motor will move from a slow speed to a faster speed. The higher the value the slower the motor accelerates. The power on default is 5. The formula for relating the acceleration to the parameter is as follows:

$$\text{Accel (steps/sec}^2\text{)} = 1,050,422 / A \text{ (parameter)}$$

The formula for determining the time during acceleration is:

$$\text{Time} = (\text{Max speed} - \text{Base speed}) / \text{Acceleration}$$

The formula used to approximate the number of steps during acceleration is as follows:

$$\text{No. of Steps} = \frac{1,050,422 * (\text{Time}^2)}{(\text{Accel parameter})} + (\text{Base speed}) * \text{time}$$

B dddd

Base rate is the speed at which the motor motion starts and stops. This value is given in pulses per second and has a range of 50 to 5000. The initial value is 1000. A verify of this register will return the actual speed as calculated by dividing the crystal frequency by four and then by the value selected as the base speed.

CH n

Hold time is a value from 0 to 127 representing the number of clock ticks to leave power on the motor after a move. This command causes the POWER pin to be pulled low within a specified number of ticks after a move of the motor is completed and it will stay low until just prior to the start of the next move command. This command

series of indexes. Without the F command any subsequent G commands received while the motor is indexing would set the "Not allowed while moving" message. Caution: If this command is used while the motor is executing a Slew command the only way to stop is with a reset or a hard limit switch input.

- G Start the motor indexing as defined by the P or N command.
- H i This command starts the motor looking for a home position in the direction last given by a +, -, or P command. It is not necessary to use a G command as motion will start immediately. If i equals zero then the motor will move at high speed until detecting a soft limit switch and then decelerate to the base speed and continue until detecting the home limit switch. If i equals one then the motor moves at high speed until the home limit switch is detected and then decelerates to the base speed reverses direction and continues until the home limit switch is not detected. If the home limit switch is not detected at the time that the motor reverses direction it is assumed to have completely passed the home limit switch and the motor will search for the switch to be detected again before looking for the not detected state. In this mode, the home limit switch should not be connected to a multiplexer for detecting direction since the SMC25 must detect the same switch in both directions. When the home position is reached the position counter is set to zero.
- I xx, yy, zz If the input pins, as defined by the mask value xx (0-255), are equal to the value of yy (0-255), then go to program location zz (0-119). This command is only valid when used in the internal program. The first value entered (xx) specifies the binary weighted decimal value of the inputs pins to be tested. The second value entered (yy) is the binary weighted decimal pattern that these selected inputs are to match. If the data and selected pins match then the next command executed will be at the program location specified by zz (0-119). If there is no match then the next instruction is the one immediately following the IF command.
- J zz Set slow jog speed. The value of zz (0-255) is used as a divisor of the base speed. The default value is 20.
- K Unused.
- L yy, zz Loop back yy times to location zz. This command permits a sequence of commands to be repeated a specified number of times. The loop command is only used in the internal stored program

mode. The first number entered specifies the number of times to loop and the second number specifies the program instruction to loop to. If yy is equal to zero then the loop is endless. When the specified number of loop backs have occurred then the program will continue with the next command. No nesting of loops is allowed.

- M dddd** Set the maximum speed of the motor. The range is 50 to 20,000 when the crystal frequency is 8 megahertz. The upper limit is proportionately lower for lower crystal frequencies. The default value on power up is 10,000 pulses per second. If the distance of a move is enough to accelerate and decelerate through the full speed range specified then the remainder of the move will be made at this Max rate until deceleration is required at the end of the move. If the distance is not large enough, then the controller will only accelerate until it is time to decelerate back to the base speed, resulting in a triangular speed-distance profile.
- N rrrr** Set the number of steps to move for the G command. The range is 0 to 16,777,215. The default value is zero.
- O zz, xx** Output the value xx to the general purpose output pins selected by the mask zz. The first number entered, zz (0-255), is the binary weighted decimal value of the output pins to be modified by the binary weighted decimal value of the second number entered, xx (0-255). See the section on inputs and outputs for additional detail.
- P rrrr** Set the direction and distance to achieve a move to absolute position rrrrr. The range of values is 0 to 16,777,215. A G command is required to effect the move. The default value is zero.
- Q** This command stops execution of the internal program if it is used in the program. If the motor is indexing it will ramp down and then stop. Use this command to stop the motor after issuing a slew command.
- R zz** Start execution of the internal program at location zz (0-119). By specifying the entry point to the internal program more than one program may be stored internally at the same time. This command is also used as an unconditional branch in the internal program.
- S** Slew the motor up to maximum speed and continue until reaching a hard limit switch or receiving a quit (Q) command.
- T** Trace one step of the internal stored program. Each time the T command is entered one command is executed from the internal

stored program. The VR command can be used to read the address of the command that will be executed next.

U xx, yy, zz Until the input pins represented by the mask xx equal the value yy branch to location zz. This command is only valid in the internal program. Refer to the IF command for a more detailed description of the parameter format.

V c Verify the contents of the register specified by the letter following the the V command. The data is sent as an ASCII decimal string followed by a carriage return and a line feed. If a Verify Enter command (VE) is sent the SMC25 returns the entire internal stored program. Each numeric parameter is followed by a carriage return and line feed. Illegal commands, such as might be found in the unused part of memory, will return the underline () character so that the host program can keep track of where everything is in memory. The verified program will be terminated by a period (.). The following is a list of the registers that can be verified.

- A Acceleration
- B Base speed
- D Deceleration
- E Internal program
- G Steps remaining in current move. Zero if not indexing.
- H Hold time
- I Input pins
- J Slow jog speed
- M Maximum speed
- N Number of steps to index
- O Output pins
- P Position. If motor is indexing this returns the position at the end of the index.
- R Internal program pointer used by trace(T) or continue(X) commands. Also updated by enter(E) command.
- W Ticks remaining on wait counter
- X Crystal frequency

W zz Wait for zz (0-255) ticks before continueing if in internal program mode, otherwise just send a message when time is up.

X Continue internal program from current position.

Y Unused.

Z rrrrr Set the position register. This command sets the internal absolute position counter to the value of rrrrr. As with other commands, if the

parameter is not entered a zero is the default value.

@c.c,

Upon receipt of the @ command all axes are deselected. For each character in the list if the character matches the address of this device then it is selected and the "Selected" output will be set low. If it does not match then the "Selected" output will be set high but the internal selection state of the device will not be changed. This is to allow external three state drivers to prevent bus contention when selecting multiple devices simultaneously. Only the most recently selected device will send data to the host. The list of characters is terminated by a comma. If just a "@," is sent then all axes are deselected. The valid address characters are 0-9 and A-V. These correspond to addresses 0-9 and 10-31.

+

Set the motor direction of movement to positive.

-

Set the motor direction of movement to negative.

%

Pole the device for any waiting messages such as errors or end of move. The responses are defined in the following table.

Description

- 0 Normal, no service required
- 1 Command error, illegal command sent
- 2 Range error, an out of range numeric parameter was sent
- 3 Command invalid while moving (e.g. G,S,H)
- 4 Command only valid in program (e.g. I,U,L)
- 5 End of move notice, a previous G command is complete
- 6 End of wait notice, a previous W command is complete
- 7 Hard limit stop, the move was stopped by the hard limit
- 8 End of program notice, internal program has completed
- G Motor is indexing and no other notice pending

?

This command returns two numbers. The first number is the part identifier which is always 25 for the SMC25. The second number is the revision number.

. (period)

If a period is received the motor will stop, the timer will be cleared, and if the internal program is executing it will stop. It is possible for this command to stop execution of the internal program in the middle of an instruction causing the parameter for that instruction to be read incorrectly. For this reason the period command is only intended as a debugging tool.

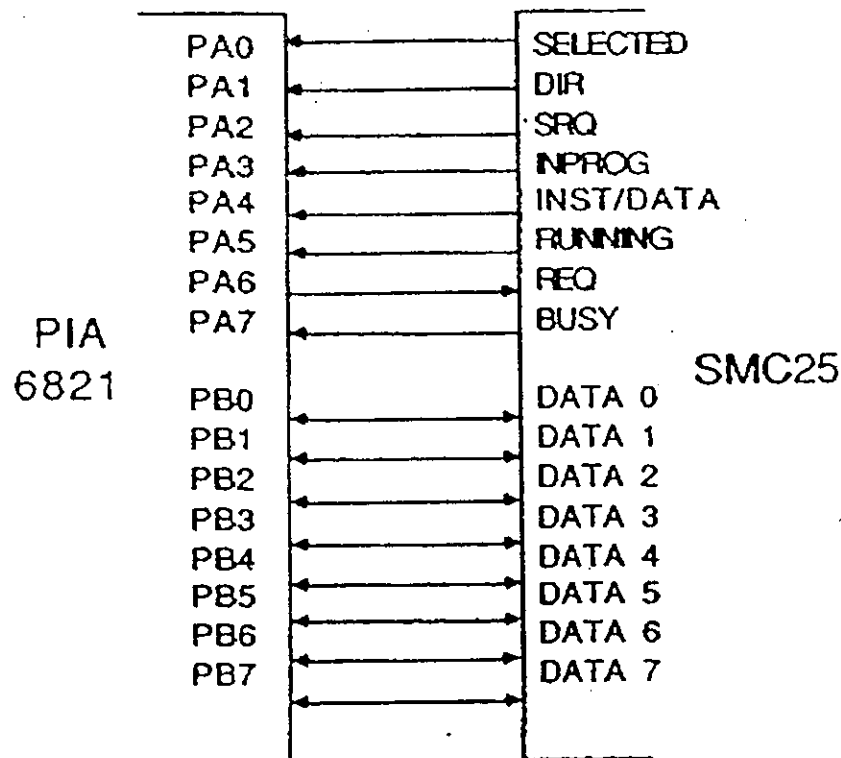
Program Memory Usage

Command/Code		<256	256-65535	≥64K
Accel	A	2	-	-
Base spd	B	2	3	-
Decel	D	2	-	-
Go	G	1	-	-
Home	H	1	-	-
If	I	4	-	-
Jog rate	J	2	-	-
Loop	L	3	-	-
Max spd	M	2	3	-
Number	N	2	3	4
Output	O	3	-	-
Position	P	2	3	4
Quit	Q	1	-	-
Jump	R	2	-	-
Until	U	4	-	-
Wait	W	2	-	-
Zero*	Z	2	3	4
Forward	+	1	-	-
Reverse	-	1	-	-

* If parameter is zero then only one byte

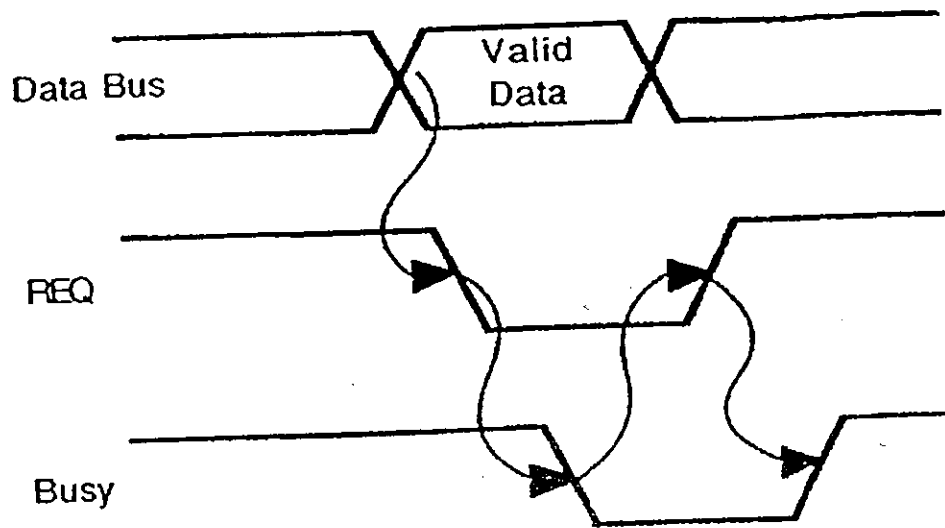
Parallel Communications

The SMC25 parallel communications capability provides the user with a high speed interface that may be connected to any of the popular microprocessor parallel port chips. Shown below is an example using the Motorola 6821 PIA.

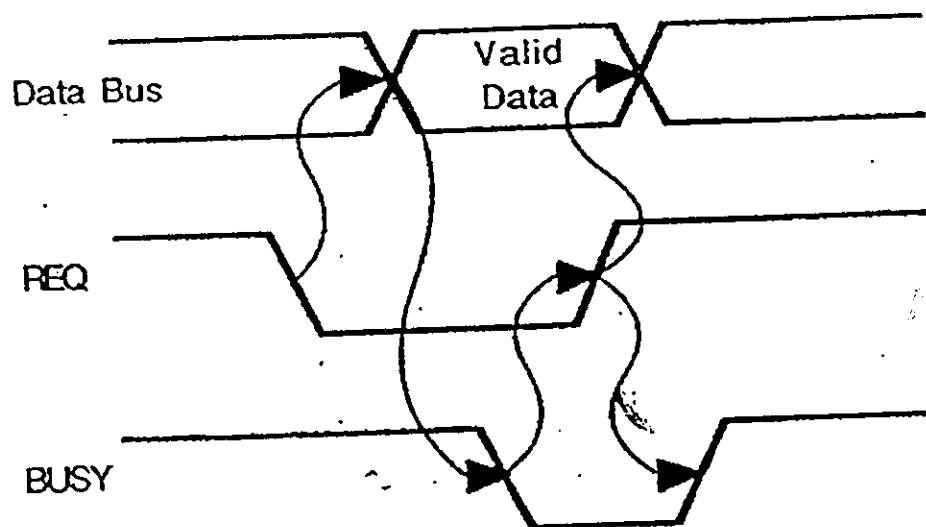


The B port of the PIA is attached to the bidirectional data bus of the SMC25. The A port is used for control and status signals. The two most important control signals are REQ and BUSY. These control the flow of data across the bidirectional data bus. REQ is driven by the host system through the PIA using port A bit 6. It is used to initiate all command and data transfers. BUSY is driven by the SMC25 and is used to let the host system know when the transfer is complete. The diagrams on the next page illustrate the relationship between these two signals. The DIR signal lets the host know which way the SMC25 expects the data to travel on the bus. Low indicates host to SMC25 and high indicates SMC25 to host.

The SRQ signal can be used as an interrupt to the host for events such as errors, limit switch stop, end of index, end of pause, or end of program. The NAND of REQ and BUSY can be used as an interrupt to signal that the SMC25 is ready for another data transfer.



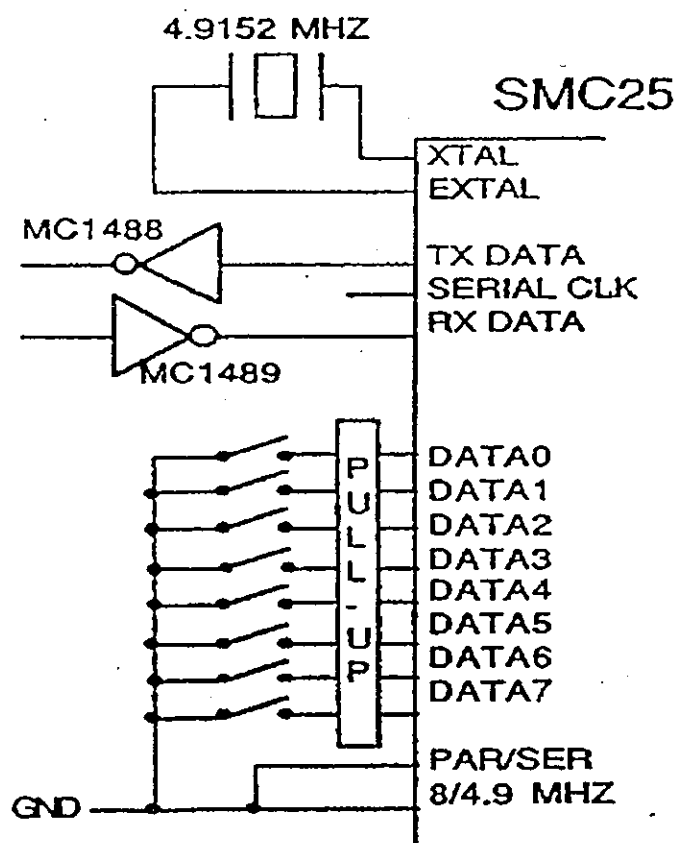
Write to SMC25



Read from SMC25

Serial Communications

The serial port provides asynchronous bit serial communications compatible with most terminals and microcomputers. Shown below is a very simple scheme for a three wire RS-232 connection with ground, transmit data, and receive data. The switches attached to data lines five through seven control the baud rate (see table) and the switches attached to data lines zero through four select the address of the SMC25.



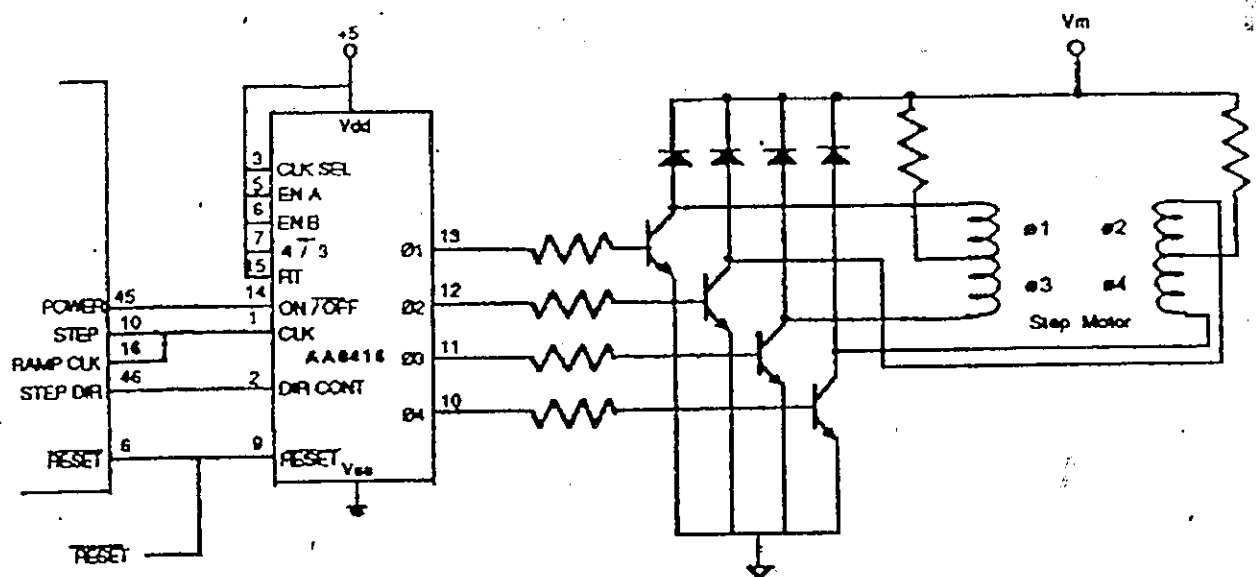
Use of the 4.9152 Mhz. baud rate crystal limits the motor speed to about 12,000 pulses per second. If your system requires speeds faster than this then an 8 Mhz. crystal may be used to allow up to 20,000 pulses per second. An external baud rate generator should then be used to provide normal transmission speeds. The external clock rate should be 16 times the desired baud rate. The communications format is 8 data bits, 1 stop bit, and no parity.

	SW7	SW6	SW5
External	OPEN	X	X
300 BAUD	CLOSED	OPEN	OPEN
1200 BAUD	CLOSED	OPEN	CLOSED
9600 BAUD	CLOSED	CLOSED	OPEN
76800 BAUD	CLOSED	CLOSED	CLOSED

4.9152 Mhz Crystal

Motor Driver Connections

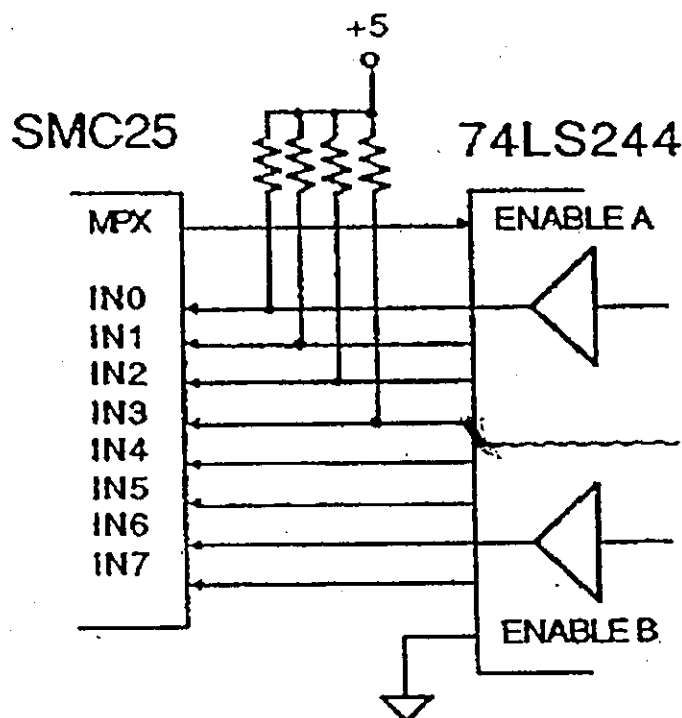
There are only three signals required to control a step motor driver system. The STEP signal controls the distance that the motor moves, STEP DIR selects the direction that the motor will turn, and POWER controls whether the motor is on or off. The drawing below shows how to connect the SMC25 to the AA8416 step motor drive control chip.



Inputs and Outputs

There are eight inputs and eight outputs provided for the user that are not generally used for other functions of the step motor controller.

The inputs can be read by the host controller with the "Verify Inputs" (VI) command which returns a number from 0 to 255. The number is the binary weighted value of the state of the inputs with IN0 being the least significant and IN7 being the most significant. During the internal program execution the inputs may be tested individually or in groups by the "If" (I) command or the "Until" (U) command. If the host interface is in the parallel mode then the SMC25 expect to see its address on IN0 to IN3 when the MPX signal is high. If there is only one SMC25 connected to the parallel interface then the address may be defaulted to "F" by the circuit shown below. The tri-state buffer also provides a level of protection from the outside world.

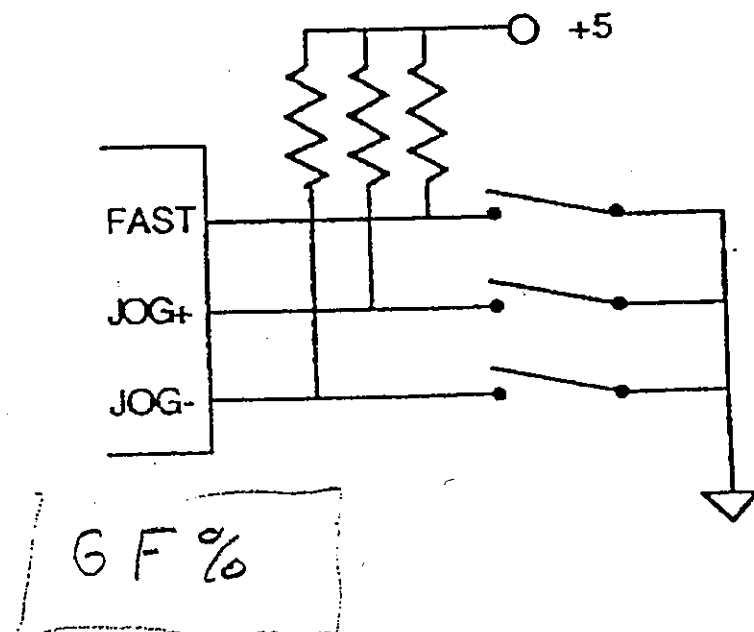


The outputs are controlled by the "Output" (O) command in either the command mode or the internal program mode. The output lines are undriven during the time that RESET is low. Shortly after the RESET line goes high the output lines will be driven low and stay that way until instructed by the user or the internal program.

Jog Switches

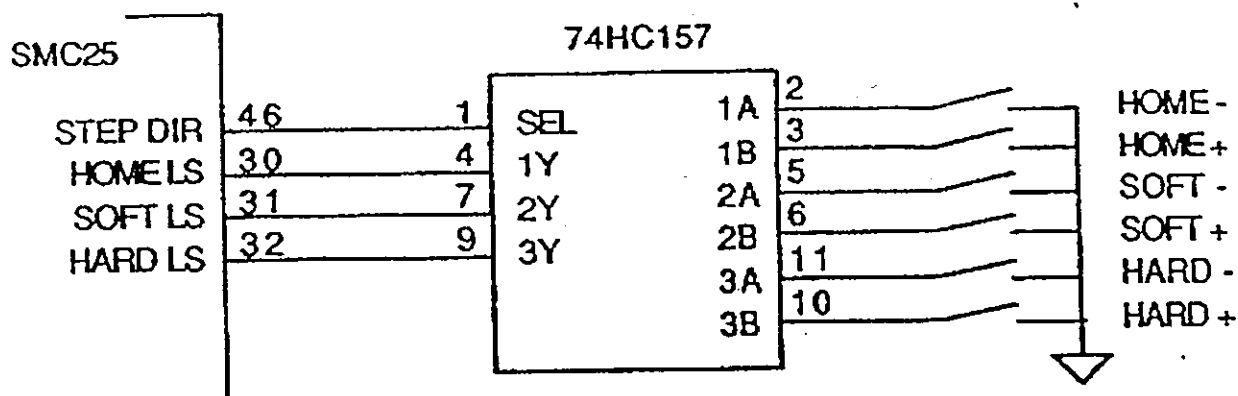
The jog switches allow the user to move the motor under manual control. These controls are only active when the motor is otherwise idle. During indexing of the motor these inputs have no effect. The JOG+ input causes the motor to move in the positive direction and the JOG- input causes the motor to move in the negative direction. These are both active low signals. If the motor is moving in response to one of the jog inputs it will continue to jog in that direction until either an index command or both jog inputs go high. Rolling from one jog input to the other without first having both high will not reverse direction. When the FAST input is high the jog inputs move the motor at the base speed divided by the jog factor. When the FAST input is low the jog inputs move the motor at the base speed. The jog inputs remain active during internal program execution as long as the motor is not indexing.

The jog inputs may be used in conjunction with a host computer to implement a learn mode of operation. The user can move the motors to the desired position with the jog inputs and then the host computer can read back the absolute position of the motors in the system and save it in a table from which it can be read back at a later time to duplicate the original motions.



Limit Switches

The limit switch inputs are used to stop the motor before the end of an index. There is one active low input for each of the three types of limit switch. These may then be connected to an external multiplexer along with the STEP DIR control line to provide separate inputs for the plus and minus directions. All three limit switch inputs should normally be high with low being the active state.



The hard limit switch (HARD LS) will unconditionally stop the motor and prevent it from further motion until the input is high again. There is no ramp provided when the hard limit is activated so the actual position of the motor may not correspond to the internal counter if the motor is moving faster than the base speed.

A soft limit switch (SOFT LS) is provided to allow accurate stopping from high speeds. When the soft limit is activated the motor will ramp down to the base speed and then proceed until there is a hard limit or the end of the index count. In this way a second limit switch can warn the system of the approaching hard limit in time to slow down to a safe speed for an accurate stop.

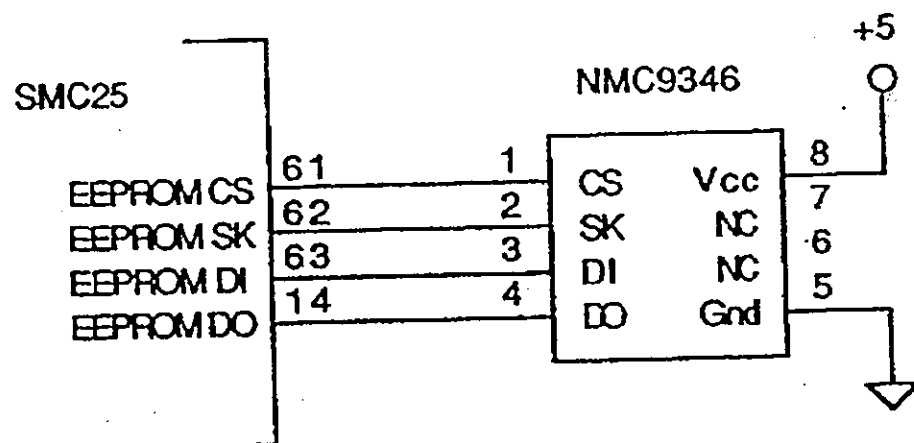
Many systems, besides limiting the travel of the motor, need to align the motor to a position somewhere between the limits. The home limit (HOME LS) is provided for this purpose and can be used in one of two ways to locate the home position. When using the H1 mode of the home command the home limit switch should not be multiplexed by the direction control since this mode needs to sense the same switch in both directions. For a complete description of these two home modes see the home command (H) description.

EEPROM Interface

The EEPROM interface allows the user to preserve the internal program and some of the registers. These are then automatically reloaded next time the system is turned on. The user may optionally specify the internal program to start executing as soon as it is loaded by providing a valid start address at the time the program is saved.

There are 128 bytes in the EEPROM of which 120 are used for the internal program. The remaining 8 are used to store the start address, the crystal frequency, the maximum speed, the acceleration factor, and the deceleration factor.

Four control lines are used to access the EEPROM. The signal names on the SMC25 are the same as the signal names on the EEPROM with an "EEPROM" on the front of each name. If there is no EEPROM installed in the system then the EEPROM DO line should be tied high and the other three lines should be left open.



EEPROM Connections

Electrical Specifications

ABSOLUTE MAXIMUM RATINGS

Item	Symbol	Value	Unit
Supply Voltage	V _{cc}	-0.3 ~ +7.0	V
Input Voltage	V _{in}	-0.3 ~ V _{cc} +0.3	V
Operating Temperature	T _{opr}	0 ~ +70	°C
Storage Temperature	T _{stg}	-55 ~ +125	°C

DC Characteristics V_{cc} = 5V ±10%, V_{ss} = 0V, T_a = 0 ~ 70 °C

Item	Symbol	Min	Max	Unit	Remarks
Input "High" Voltage	V _{ih}	2.2	V _{cc} +0.3	V	--
Input "Low" Voltage	V _{il}	-0.3	0.8	V	--
Output "High" Voltage	V _{oh}	2.4	--	V	I _{oh} = 0.2mA
Output "Low" Voltage	V _{ol}	--	0.4	V	I _{ol} = 1.6 mA
Supply Current	I _{cc}	7.0	20.0	mA	--

AC Characteristics V_{cc} = 5V ±10%, V_{ss} = 0V, T_a = 0 ~ 70 °C

Item	Symbol	Min	Max	Unit
Cycle Time(EXTAL/4)	T _{cyc}	0.5	1.0	μs
Crystal Frequency	T _{xfr}	4.0	8.0	MHz
Oscillator Stabilization Time	T _{rc}	20	--	ms
Reset Pulse Width	PW _{rst}	3	--	T _{cyc}
Step Clock Pulse Width	T _{scw}	36	75	T _{cyc}
Step Clock Period	T _{scp}	100	40,000	T _{cyc}

Physical Dimensions

(Unit:mm)

